Water Quality Prediction

Problem & Scope

Predicting the CCME Water Quality Index: tree ensembles vs a neural model

Phase 1: Naive baseline, Random Forest, XGBoost | CSAI (AML / DL) Project 18

- Goal: predict the continuous water quality index (**CCME_Values**) from routine physical and chemical measurements
- Task type: **regression** (observed range 31.304 to 100.0)
- Phase 1 compares a Naive baseline, Random Forest and XGBoost on one shared, leakage-safe split
- Phase 2 (planned) adds a regularized MLP (PyTorch) on the same split and preprocessing

Phase 1 scope. All three models are scored on the **same** test set via the same **evaluate()** call. Both tree models expose the same training-function signature, so the Phase 2 MLP plugs into the identical evaluation path.

2,827,977
rows in the raw dataset (14 columns)

5
countries: England, USA, Ireland, China, Canada

61,689
unique monitoring sites (Area) in the raw data

Dataset and Data Cleaning

Data Analysis

Harmonized global surface-water monitoring dataset, 5 countries

Rows by country (raw data)

Country	Rows
England	2,129,198
USA	413,814
Ireland	235,019
China	45,997
Canada	3,949

Data handling rules

- Units verified per column; no mixed-unit rows found
- No detection limits present (resolved upstream)
- 5 of 8 measurement columns carry their exact median at 11 to 32% of rows: flagged, not inherited silently
- Impossible values (pH outside 0 to 14, negative concentrations, temperature above 40 °C) flagged, not deleted
- 9,971 exact duplicate rows (0.35%) removed
- Imputation is not done during cleaning: it lives in the preprocessing pipeline, fitted on train only

2,818,006
rows x 30 columns after cleaning

85.05
mean CCME (median 90.60)

17.65
standard deviation of the target

45 MB
clean.parquet, committed (plain git, no LFS)

Target and Leakage

CCME Water Quality Index (CCME_Values)

Target Definition

$CCME_Values = 100 - sqrt(F1^2 + F2^2 + F3^2) / 1.732$

- **F1 (Scope):** percentage of variables that do not meet their guideline
- **F2 (Frequency):** percentage of tests that do not meet guidelines (set equal to F1 here)
- **F3 (Amplitude):** amount by which failed values exceed their guidelines
- Dividing by 1.732 (square root of 3) normalizes the maximum distance, 173.2, to 100

Index bands

95 - 100	Excellent
80 - 94	Good
65 - 79	Fair
45 - 64	Marginal
0 - 44	Poor

Leakage decisions

- **CCME_WQI** is excluded: it is the target cut into 5 classes, so it is direct label leakage
- The 8 chemical parameters (Ammonia, BOD, Dissolved Oxygen, Orthophosphate, pH, Temperature, Nitrogen, Nitrate) fully determine the index (0.000000 reconstruction error)
- So the models are a **surrogate** of the CCME formula, not a discovery of water physics
- **Area** and raw **Date** are excluded from the features; only sin_day, cos_day and month are kept

Why it matters. The target is a function of the input measurements, so the 8/8 results are labelled **surrogate_emulator_benchmark**: formula approximation, not independent prediction. Partial-measurement scenarios (7/8, 6/8, 4/8) test the machine learning utility.

Method and Fairness Rules

Pipeline & Models

One shared split, train-only preprocessing, test set touched once

1 Raw data | 2 Clean | 3 Shared split | 4 Preprocess | 5 Train 3 models | 6 Evaluate once

Fairness rules (every model)

- One split, **SEED = 42**, assignments saved to disk
- Split by **Area** (monitoring station), stratified by Country: no station appears in two partitions
- Preprocessing and aggregates fitted on train only
- Tuning on validation only
- Test set used exactly once

Shared split (70 / 15 / 15)

Partition	Rows	Stations
Train	1,986,397	43,382
Validation	415,805	9,153
Test	415,804	9,154

China: all 45,977 rows come from one station (Hou Bay), allocated to train, so China has 0 test samples.

Models

- **Naive baseline:** median predictor, the floor to beat
- **Random Forest:** search over trees, depth, min leaf size, features per split
- **XGBoost:** Optuna search with early stopping

MAE
mean absolute error

RMSE
penalizes large errors more

R^2
variance explained

Results, Team and Next Steps

Evaluation & Docs

Test set, same split, same evaluate() call

Test results

Model	MAE	RMSE	R^2
Naive baseline (median)	13.6047	19.4916	-0.0923
Random Forest (100k rows)	0.4078	0.9197	0.9976
XGBoost (full, engineered)	0.1000	0.3282	0.9997

Ablation (XGBoost, test)

Configuration	MAE	RMSE
Raw features (10)	0.1517	0.4005
Raw + engineered (23)	0.0999	0.3282

Partial measurements (test R^2)

Measurements	Random Forest	XGBoost
8/8	0.9976	0.9997
7/8	0.8093	0.8341
6/8	0.7901	0.8139
4/8	0.5304	0.5728

SHAP. Orthophosphate, log_Orthophosphate and Ammonia carry the largest mean absolute SHAP values.

Phase 2 (planned)

- Regularized MLP in PyTorch: dropout, weight decay, early stopping, batch norm
- Ablations on depth, width and regularization; optional MLP-on-residuals hybrid on top of XGBoost

Team Tajuddin: data and EDA | Parthraj: target, split, evaluation | Himanshu: features, Random Forest | Jivit: XGBoost | Satya: metrics, baseline, report and deck